When Do Redistribution Mechanisms Matter? Behavioral Heterogeneity and Group Composition in Common-Pool Resource Dilemmas

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March 31, 2026

Abstract

This paper examines whether the redistribution mechanisms proposed by Koster et al. (2025) [8] remain equally effective when applied to populations with real behavioral heterogeneity rather than simulated agents. Using experimental data from public goods games (362 subjects), we identify behavioral clusters and train LSTM models that act as behavioral clones. These models are used to simulate repeated interactions under three redistribution mechanisms: Equal, Proportional, and Interpolating. Our results partially replicate the main findings of Koster et al. [8], as the interpolating mechanism continues to provide a balanced trade-off between efficiency and inequality. However, its advantage in efficiency over the proportional mechanism is reduced and no longer statistically significant. Furthermore, we show that the performance of these mechanisms strongly depends on group composition, with differences becoming particularly pronounced in heterogeneous and asymmetric groups. In general, our findings highlight that behavioral heterogeneity is a key factor in understanding the effectiveness of redistribution mechanisms and suggest that their performance is context-dependent rather than universal.

1 Introduction

The social dilemmas associated with common-pool resources have been widely studied in economics and social sciences, as they highlight the tension between individual interest and collective welfare. In this type of context, individual interest and collective interest do not always coincide, which can lead to actions that result in the deterioration of the shared resource. Since the publication of "The Tragedy of the Commons" by Hardin (1968) [6], an important part of the literature has sought to understand under what conditions this type of dilemma can be resolved and which mechanisms allow individual incentives to be better aligned with group objectives.

Following this line, recent research has focused on analyzing how different redistribution mechanisms affect individual behavior. In particular, Koster et al. (2025)[8] introduce an innovative approach based on reinforcement learning to design adaptive mechanisms that balance efficiency and equity. Their results show that such mechanisms can outperform traditional ones such as egalitarian and proportional redistribution.

However, their study relies on simulated agents that imitate human behavior. This raises the question of whether these results hold when real behavioral heterogeneity is introduced. Additionally, we explore whether the performance of these mechanisms depends on group composition.

To address these questions, we use experimental data from public goods games to identify behavioral types and simulate interactions under different redistribution mechanisms. Our results show that while the general pattern identified by Koster et al.(2025) [8] holds, behavioral heterogeneity reduces differences in efficiency and makes outcomes dependent on group composition.

2 Theoretical framework

2.1 Social dilemmas and cooperation

The literature on common-pool resources (CPR) starts from the idea that there are certain resources whose collective management generates tensions between individual interest and group welfare. This problem was popularized by Garrett Hardin in his well-known essay called the "tragedy of the commons" [6], where he argues that, in open-access contexts, individuals have incentives to overexploit the resource, even when this leads to its depletion.

Hardin's argument is based on the main idea that each individual obtains the full benefit of using an additional unit of the resource, while the costs of that use are shared among all. Therefore, from the individual point of view, it is rational to continue using the resource more and more, even if this ends up harming the group as a whole [**?**].

These types of situations are not only theoretical approaches, but we can observe them in numerous real contexts, such as overfishing, deforestation, pollution or the use of shared infrastructures. In all these cases two common characteristics appear: on the one hand, it is difficult to prevent individuals from accessing the resource and, on the other hand, its consumption is rival, so that the use by one person reduces what remains available for others [12].

This combination generates a clear incentive problem, which is that when individuals make decisions independently and guided by their own interest, the aggregate result tends to be inefficient, that is, the resource is overused or less than necessary is invested in its maintenance. From the perspective of game theory, this implies that, in situations of one-shot interaction, agents have incentives to overexploit the common resource [12].

These dynamics can be understood as strategic dilemmas, similar to the prisoner's dilemma, where the rational strategy at the individual level does not coincide with the best solution for the rest. Each individual knows that moderating their behavior or investing in the care of the resource would benefit the group, but the costs would be borne directly, while the benefits would be shared among all. These problems can be framed within the so-called social dilemmas, in which rational decisions at the individual level lead to inefficient outcomes from the group's perspective [13].

However, as Gardner, Ostrom and Walker (1990) [4] point out, not all common-pool resources necessarily lead to this type of outcomes. The emergence of the dilemma depends on specific conditions being met, among them the subtractability of the resource and the difficulty of excluding others from it. When these conditions intensify, incentives for non-cooperative behavior increase and, with it, the risk of overexploitation of the resources in question.

Faced with Hardin's pessimistic diagnosis, which proposed solutions based on privatization or state control, the subsequent literature has shown that individuals can, under certain conditions, develop sustainable cooperation strategies. In particular, the work of Elinor Ostrom shows that communities can self-organize to manage common resources through shared rules, monitoring mechanisms and sanctions, thus being able to avoid overexploitation [11].

Therefore, based on Ostrom's work (1990) [11] we can conclude that the dilemmas associated with common-pool resources are not inevitable, however, it is necessary to find mechanisms that align individual incentives with collective objectives. And it is precisely this need that has motivated a large part of the subsequent literature. Understanding how these problems arise and under what conditions they can be solved is understood as something vital to analyze the design of institutions and allocation mechanisms, issues that we will address in the next section of this brief literature review.

2.2 Redistribution mechanisms and incentives

An important part of the literature on common pool resources dilemmas has focused on analyzing how different allocation mechanisms influence individuals' behavior. In many cases of these common-pool resource dilemmas, there is some type of rule, not necessarily explicit, that

determines how benefits or access to the resource are distributed, and these rules directly condition incentives to cooperate.

Two of the most basic mechanisms that have been studied are equal allocation and proportional allocation. In the first case, all individuals receive the same amount of resources regardless of how they decide to act. The problem with this type of distribution is that if the outcome is the same for everyone, some individuals may choose not to contribute and benefit from the contribution of others. This free-riding phenomenon has been widely documented in public goods experiments, where cooperation tends to deteriorate when there are no mechanisms to penalize opportunistic individuals [3].

On the other hand, proportional mechanisms introduce a more direct relationship between contribution and reward. In these cases, those individuals who have contributed more receive a larger share of the resource, which can encourage more cooperative behavior. However, this type of system also has its limitations, as it can generate dynamics of persistent inequality. For example, in the case of individuals who start from a disadvantaged position, they may become trapped without the ability to improve their position [2].

This contrast reflects a classic trade-off in mechanism design, which consists in that those that generate greater total surplus tend to produce higher levels of inequality, while more egalitarian mechanisms tend to reduce incentives to contribute and, therefore, the total efficiency of the group [1]. In addition, experimental evidence suggests that individuals do not respond only to monetary incentives, but also to other factors such as fairness and reciprocity. When a mechanism is perceived as unfair, cooperation may deteriorate even if the system is efficient [2].

Apart from these two mechanisms, later studies have also focused on analyzing other more complex forms of distribution based on cooperation among the individuals themselves. From cooperative game theory, different criteria have been proposed to divide benefits (such as, for example, the Shapley value, the core or the nucleolus) that have sought for the distribution to be, at the same time, efficient and as fair as possible [10]. However, a distribution being "fair" from a theoretical point of view does not mean that it always works in practice. Some individuals may not agree with the outcome or feel that they are disadvantaged, which may lead them to deviate. For this reason, in addition to efficiency and equity, the stability and functioning of the mechanism has continued to be considered a key aspect of study.

The reviewed results from previous studies have led us to conclude that the design of the allocation mechanism is a central element to understand cooperation in common resource contexts. However, most of the mechanisms analyzed in this literature are static, in the sense that they do not adapt to the evolution of agents' behavior or to changes in the system. Therefore, this limitation opens the door to more dynamic approaches, such as those we proceed to analyze below.

2.3 Adaptive mechanisms and artificial intelligence

More recently, works have appeared that use artificial intelligence to design more flexible allocation mechanisms. Koster et al. (2025) [8] decided to use reinforcement learning to train a "social planner" capable of allocating resources dynamically. The idea behind this paper was that the mechanism would not follow a fixed rule, but rather that it would adapt its decisions based on what had previously occurred and the state of the resource.

This study also connects with a broader line of research that emphasizes that, in real-world social dilemmas, cooperation and defection are not one-shot decisions, but properties of policies that develop over time[9]. For this reason, it was shown that sequential models such as those using multi-agent learning allow better capture of the temporal structure of these problems than more static ones, such as those we have analyzed previously.

The results of Koster et al. (2025) [8] showed us that this type of mechanism can generate more total surplus than traditional mechanisms and, at the same time, can avoid extreme levels of inequality. In this case, it was observed that the learned agent tends to be more egalitarian

when the resource is abundant and more selective when it is scarce, which helps sustain cooperation. In turn, other works have also shown that, if agents are given a preference for resulting equity among individuals, their behavior changes and they can end up cooperating more, since they do not act only to maximize their own benefit (Hughes et al., 2018).

However, these results are obtained in most cases using simulated agents that imitate human behavior. This implies that although in this way it is possible to study individuals' behavior under these new mechanisms, it is not clear to what extent the results obtained with trained agents hold when the mechanism is applied to real populations, where human behavior is more diverse and less predictable.

2.4 Behavioral heterogeneity

As we mentioned in the previous section, one of the main limitations of many studies that use simulated agents is that they simplify individuals' behavior, so they do not fully capture the diversity present in real populations. In this study, we consider that taking this heterogeneity into account is important because the outcome of a mechanism can depend heavily on the composition of the group. While the same mechanism may work well in a group with many cooperators, it may fail in another with more free-riders.

Despite this, many models simplify human behavior and do not fully capture this diversity. As a consequence, they may overestimate the effectiveness of certain mechanisms when transferred to real contexts. In their study, Koster et al. (2025) [8] show that allocation mechanisms are key to sustaining cooperation and that adaptive mechanisms, in this case their interpolated mechanism, achieve a good balance between efficiency and equity, generating high levels of surplus without producing extreme inequalities. However, the study is based on simulated agents, which simplifies the diversity of human behavior and does not allow for a proper analysis of how this affects the performance of the mechanisms.

This study precisely addresses this issue by replicating and revising the analysis of Koster et al. [8] using real human behavior data. Instead of assuming homogeneous agents, we identify different behavioral profiles and use these patterns to build more realistic behavioral models. Based on these models, we simulate heterogeneous populations and analyze how the composition of groups influences outcomes. In this way, we extend their approach by incorporating realistic behavioral heterogeneity and studying its impact on efficiency, inequality and the sustainability of the mechanisms.

3 Research Question and Hypotheses

Based on the limitations identified in the previous literature review, in this third section we pose our two research questions:

RQ1: Do the results obtained by Koster et al [8] . regarding the performance of redistribution mechanisms hold when they are evaluated in populations that incorporate behavioral heterogeneity based on real data, instead of simulated agents?

RQ2: Does the performance of allocation mechanisms depend on the behavioral composition of the groups, in terms of greater or lesser heterogeneity among their members?

To answer the proposed questions, we formulate the following hypotheses:

H1: The introduction of behavioral heterogeneity based on real data modifies the balance between efficiency and inequality observed in Koster et al., which affects the relative performance of redistribution mechanisms.

H2: The composition of the groups influences the results of the mechanisms, such that more heterogeneous groups may generate different outcomes than more homogeneous groups.

4 Data and Methodology

4.1 Data

To carry out our analysis we have used a dataset of human behavior obtained from public goods experiments (public goods games). Our dataset includes contribution decisions made by 362 participants over 10 repeated rounds. In each of

these rounds, individuals decide how much to contribute to a common resource, which allows us to observe how their cooperation strategies evolve over time.

The dataset that we will use allows us to identify persistent patterns of behavior among participants and observe the differences in contribution decisions, which already indicates the existence of behavioral heterogeneity in the population. This heterogeneity is especially relevant for the extension of the study by Koster et al. (2025) [8] that we will carry out, as it will allow us to construct differentiated behavioral profiles and subsequently use them in the analysis of allocation mechanisms.

4.2 Methodology

Behavioral clustering and behavior modeling

The first step of our analysis will consist of representing heterogeneity in human behavior. To do this, we will group participants into behavioral clusters based on their contribution trajectories in the public goods games from which we have obtained our data. With this grouping we aim to reflect persistent differences in cooperation patterns over time.

Based on these clusters, we will train a predictive model of behavior for each group; more specifically, what we will do is estimate a recurrent neural network of the LSTM type for each cluster, with the objective of predicting how individuals contribute. Our model takes as input the history of behavior (own contributions, the average of others, the current round and the endowment) and, from there, predicts the level of contribution (between 0 and 1). Thus, each model acts as a behavioral clone of a specific type of participant. In this way, instead of working with representative agents, we use these models to construct populations that reflect the behavioral heterogeneity observed in the data.

Allocation mechanisms

Once the behavioral models are trained, we use them to simulate repeated interactions under three redistribution mechanisms taken from Koster et al. The first of these mechanisms is the Equal mechanism, in which the available resource is distributed equally among all players. The second is the Proportional mechanism, in which allocations depend on each player's contribution in the previous round. And finally, we use the Interpolating mechanism, which combines proportional and equal redistribution depending on the size of the common resource. In this case, the mechanism becomes more proportional when resources are scarce and more egalitarian when the fund is abundant. In our implementation, we maintain the same functional form and the same parametrization proposed by Koster et al. [8] , with k=22.

Simulation design

The game we simulate follows the structure of a repeated common-pool resource problem, that is, in each round, the mechanism allocates resources to four players from a shared pool, and these decide how much to return to the pool and how much to keep as surplus. The players' contributions are what allow the resource to regenerate, implying, in turn, that if there is low reciprocity this resource weakens over time and may reach collapse.

To form the groups, we randomly select four behavioral clones according to the distribution of clusters observed in the data. From there, we simulate a large number of games under each mechanism. The simulation unfolds over 40 rounds, following the structure used by Koster et al. Following that structure, in each game we record the evolution of the pool, the allocations, the contributions, the surplus and the number of active players.

Evaluation metrics

To compare the different mechanisms, in our analysis, just as Koster et al. did, we focus on three main dimensions. First, we measure efficiency through total surplus. Second, we measure inequality with the Gini coefficient applied to the final surplus of the players. Finally, we evaluate sustainability based on the number of rounds played and by observing whether the game is sustained until the end.

Additionally, we should mention that we also consider the average number of active players

and the evolution of the common pool over time as complementary indicators of inclusion and dynamic stability.

Analysis of group composition

Finally, we decided to conclude this work by analyzing whether the performance of the mechanisms depends not only on the existence of heterogeneity, but also on how the different behavioral types are combined within each group. We decided to construct specific group compositions instead of limiting ourselves to observing only how random mixing works.

Among the different compositions that we have been able to evaluate, we find homogeneous groups, balanced mixed groups, fully heterogeneous groups and asymmetric groups, that is, groups in which one behavioral type dominates over the others. Given that clusters 4 and 5 are both highly cooperative but have few observations, we decided to group them in this part of the analysis into a single category.

For each composition, we again simulate 50 games under each redistribution mechanism. This allows us to see whether some mechanisms perform better or worse depending on how the group is formed, and thus verify whether the interpolating mechanism maintains good performance under different combinations.

5 Results

5.1 Performance of redistribution mechanisms

First, we begin by analyzing the performance of the three redistribution mechanisms (Equal, Proportional and Interpolating) using our populations constructed from behavioral clusters. Throughout the 200 simulations per mechanism carried out, we observe that clear differences emerge both in terms of efficiency and inequality. On the one hand, we observe how the Interpolating mechanism reaches the highest average total surplus (810.6), followed by the Proportional mechanism (637.5), while the Equal mechanism shows a significantly lower performance (411.1).

On the other hand, we must also point out that these gains in terms of efficiency are accompanied in turn by other differences in inequality. The Equal mechanism generates the lowest inequality (Gini = 0.075), while the Proportional mechanism produces the highest (Gini = 0.225). However, it is the Interpolating mechanism that is located at an intermediate point (Gini = 0.174), managing to reduce part of the inequality generated by proportional allocation, while still maintaining high levels of efficiency.

Therefore, this section reproduces, in general terms, the central result of Koster et al. (2025) [8] , since the interpolated mechanism once again stands out as the one that best balances efficiency and inequality. In their study, the proportional mechanism generated more surplus than the egalitarian one, but at the cost of clearly higher inequality, while the mechanism learned by RL, and subsequently the interpolated mechanism designed from it, managed to combine high levels of surplus with more moderate inequality, just as happens in our review.

However, it is worth highlighting an important difference with respect to Koster et al. [8] , in their experiment the mechanism learned by RL clearly outperformed the proportional one in terms of surplus, in our replication the interpolated mechanism obtains a higher average surplus (810.6 compared to 637.5), but this difference is not statistically significant. To evaluate these differences, we carried out Wilcoxon rank-sum tests between mechanisms and observed that the differences in total surplus are statistically significant when comparing Equal with Proportional (z = -9.23, p ¡ 0.001) and with Interpolating (z = -9.25, p ¡ 0.001). However, the difference between Proportional and Interpolating is not statistically significant (p = 0.17). This indicates that, when introducing behavioral heterogeneity based on real data, the advantage of the interpolated mechanism in efficiency is maintained, although it is smaller and not statistically significant.

In contrast, after carrying out the Wilcoxon rank-sum tests, we were also able to observe that the differences in inequality are significant in all comparisons. The Proportional mechanism generates significantly higher levels of inequality than Equal and Interpolating, while Interpolating also produces more inequality than Equal (p

¡ 0.01).

Finally, we consider it important to highlight that, in our case, the three mechanisms reach "full sustainability", since all simulations reach the maximum number of rounds. Unlike the study by Koster et al. [8] , where an important part of the comparison between mechanisms depended on whether the games managed to be sustained until the end and on how many players remained active, in our simulations all games reach the 40 rounds. Therefore, in our case the differences between mechanisms are not observed in the formal duration of the game, but in the level of resource, surplus and inequality that they manage to sustain over time.

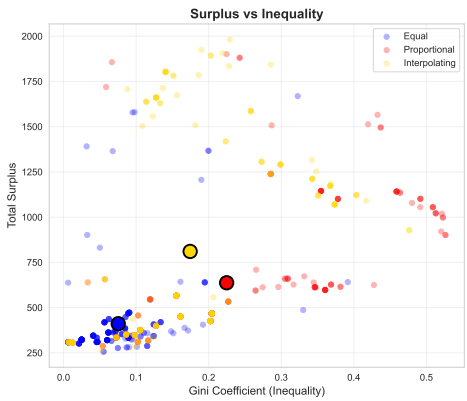

Figure 1: Relationship between efficiency (total surplus) and inequality (Gini coefficient) across redistribution mechanisms.

In this figure, we observe a clear trade-off between efficiency and equity, as the proportional mechanism increases surplus, but also inequality, while the interpolated mechanism maintains high levels of efficiency without reaching such high levels of inequality.

5.2 Dynamics of resource use

To better understand how these differences arise, we also analyze the dynamics of the common pool and contributions over time. In this step we find that in all mechanisms, the resource decreases rapidly in the first rounds before stabilizing at lower levels. However, it can be seen in the following graph how under the Equal mechanism, the pool quickly collapses to values close to zero, while the Proportional and Interpolating mechanisms manage to maintain more positive levels of the resource throughout the game.

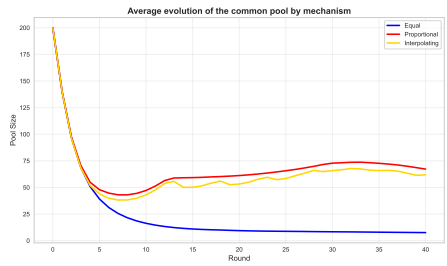

Figure 2: Average evolution of the common pool over time by redistribution mechanism.

This suggests that mechanisms that incorporate incentives (Proportional and Interpolating) are more capable of sustaining contributions over time, avoiding the rapid collapse observed under the egalitarian mechanism.

These findings partially coincide with those shown by Koster et al. [8] ; therefore, in this sense, our replication confirms that the type of mechanism affects not only the final outcome, but also the way in which the common pool evolves over the course of the game.

However, in our simulations the same pattern of player exclusion observed by Koster et al. [8] does not appear. In their study, when testing the proportional mechanism, some players stopped receiving resources and, in practice, were left out of the game. In contrast, in our case all four players remain active in all simulations. For this reason, and following what was noted earlier, in our case the differences between mechanisms are reflected in the level of resources and surplus that they manage to sustain over time

5.3 Behavioral heterogeneity

Regarding contribution patterns, we have been able to observe that the use of real behavioral clusters also introduces considerable heterogeneity in these patterns. The identified clusters range from individuals who practically do not contribute (mean ≈ 0.03) to individuals who are almost fully cooperative (mean ≈ 0.99). This heterogeneity translates into different levels of response to the behavior of others; while some clusters show strong conditional cooperation (pos-

itive slope), it is observed that others barely contribute regardless of what others do.

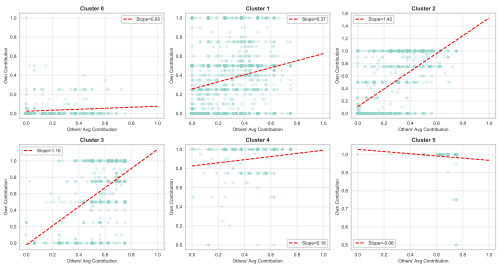

Figure 3: Relationship between individual contribution and the average contribution of others across behavioral clusters.

These results confirm that individuals do not behave as representative agents. Instead, they follow differentiated behavioral rules, which suggests that evaluating mechanisms under assumptions of homogeneity may conceal relevant results for the analysis of their performance.

In that sense, Figure 3 not only describes individual variation, but shows that subjects can be grouped into clearly distinct behavioral types, ranging from almost non-cooperative profiles to nearly fully cooperative profiles. This diversity does not appear explicitly in the presentation of results by Koster et al. [8] , and constitutes the basis of our review of the problem.

5.4 Group composition and performance

Aditionally, we analyze the role of group composition in the results obtained. Our results show that composition has an important impact on the surplus generated, although this effect is not uniform and depends on the type of group considered.

First, when analyzing specific compositions, we observe that in homogeneous groups the outcome is mainly determined by the behavioral type that composes them, and not so much by the mechanism applied. For example, groups composed exclusively of highly cooperative individuals achieve the highest surplus (1463.5) under all three mechanisms, while groups composed only of individuals from cluster 0 present much lower levels of surplus (291.3), also without relevant differences between mechanisms. This indicates that, when behavior is homogeneous, the scope for action of the mechanism is limited.

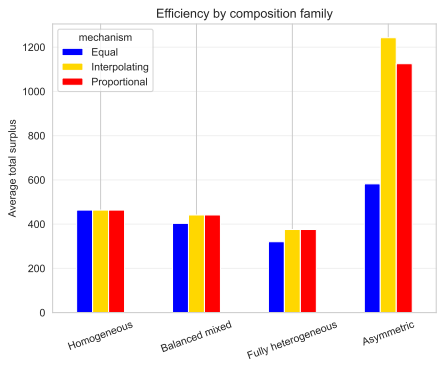

Figure 4: Average total surplus by group composition and redistribution mechanism.

The figure confirms that the effect of the mechanisms is not uniform, but rather depends on the composition of the group. In particular, the mechanisms are especially relevant in asymmetric groups, where the interpolated mechanism stands out for its ability to maintain high levels of efficiency in the presence of behavioral heterogeneity.

However, this pattern changes when we introduce heterogeneity. As shown in Figure 4, when grouping compositions by types of individuals (homogeneous, mixed, and asymmetric), the differences between mechanisms are small in homogeneous groups and relatively moderate in mixed groups, but they clearly widen in asymmetric groups. In these groups, where highly cooperative and less cooperative profiles coexist, the choice of mechanism becomes clearly decisive.

This result can be clearly observed in specific compositions. For example, in the composition “3 high cooperators + 1 low”, the Interpolating mechanism reaches a total surplus of 1689.4, compared to 1429.7 under Proportional and only 387.2 under Equal. A similar pattern appears in other heterogeneous configurations: in the composition “2 C0 + 2 High”, the surplus increases from 317.1 under Equal to 1274.5 under Proportional and 1398.5 under Interpolating; and in “All different incl. High”, from 447.4 under Equal to 1135.4 and 1290.9 respectively. Overall, these results show that the advantage of the interpolated mechanism is not general, but rather appears especially clearly when the group

combines very different profiles.

5.5 Efficiency–inequality trade-off by composition

Finally, we analyze how the relationship between efficiency and inequality varies depending on group composition. As in the case of surplus, this trade-off is not uniform.

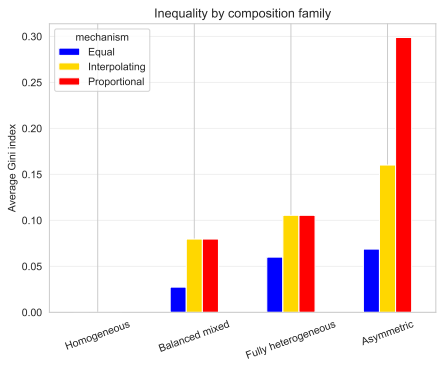

Figure 5: Inequality (Gini index) by group composition and redistribution mechanism.

This shows us that the ability of a mechanism to balance efficiency and equity depends not only on its design, but also on the composition of the group.

In homogeneous groups, levels of inequality are low and similar across mechanisms, reflecting the absence of behavioral differences within the group. However, in heterogeneous groups and, especially, in asymmetric groups, clear differences emerge. In these cases, increases in efficiency are usually accompanied by increases in inequality, particularly under the Proportional mechanism, which reaches very high Gini values (up to 0.46 in some cases).

In contrast, when considering the Interpolating mechanism, we observe that high levels of efficiency can be maintained while preserving more moderate levels of inequality. This indicates that, in contexts where there are marked differences between individuals, mechanisms not only affect the level of surplus generated, but also how these outcomes are distributed. Therefore, in contrast to the evidence presented by Koster et al. (2025) [8], our analysis suggests that the balance between efficiency and equity is not uniform, but rather depends on the behavioral composition of the group.

6 Discussion

In this section, we discuss the main conclusions that can be drawn from the results obtained in this work. First, we discuss how our review reproduces, in general terms, although with relevant nuances, the main results of the paper by Koster et al. (2025) [8]. Second, we discuss what we consider our main finding or contribution, that is, the relevance of the behavioral composition of groups and the non-uniform performance of the mechanisms tested in this study.

6.1 Robustness of the results in the presence of behavioral heterogeneity

Our first research question asked whether the results of Koster et al. would hold when the mechanisms were evaluated in populations with behavioral heterogeneity based on real data. Let us recall that, in the original work, the authors use data from human participants to train models that imitate their behavior (behavioral clones), on which they subsequently evaluate the different mechanisms.

By introducing real behavioral heterogeneity, our results indicate that, in general terms, the main pattern found by Koster et al. [8] in their study is reproduced. We observe that the egalitarian mechanism generates low inequality, but also low levels of surplus, that the proportional mechanism improves efficiency, but at the cost of significantly higher inequality, and that the interpolated mechanism manages to position itself between both, maintaining high levels of efficiency without reaching the very high levels of inequality of the proportional one.

These findings lead us to only partially accept our first hypothesis (H1), since, although in our results we observe smaller differences in terms of efficiency between the three mechanisms, we still observe better performance in the case of the interpolated mechanism. In fact, the most important difference with respect to the work of Koster et al.[8] appears precisely here, since, in their study, the mechanism learned by RL and the interpolated mechanism derived from

it clearly outperformed the proportional mechanism in terms of surplus. In our case, the interpolated mechanism also obtains a higher average surplus than the proportional one, but that difference is not statistically significant.

A possible explanation for this smaller advantage of the interpolated mechanism is that, in an environment with greater behavioral diversity, the mechanism faces more varied responses. Therefore, when players do not react in a similar way to incentives, it may be more difficult for the same rule to generate consistent improvements across all groups. By this we do not mean that the effectiveness of the interpolated mechanism is invalidated, but rather that its performance depends to some extent on how individuals behave within the group.

Beyond pointing out that the main finding highlighted by the authors holds, we also consider it important to explain and argue why this occurs. Both in the work of Koster et al. [8] and in our review, non-adaptive mechanisms present clear limitations. On the one hand, the egalitarian mechanism distributes resources without conditioning allocation on the prior behavior of the players, favoring inclusion, but weakening incentives to contribute. In the case of the proportional mechanism, it clearly rewards those who contribute more, which strengthens incentives, but also tends to reinforce differences between players, leading to very high levels of inequality.

Finally, we see how the interpolated mechanism works better precisely because it partially avoids the limitations of the previous mechanisms. When the resource is scarce, this mechanism introduces a logic closer to proportionality, and when the fund is abundant, it adopts a more egalitarian logic, which reduces inequality and avoids overly rigid exclusion dynamics. In this sense, the interpolated mechanism manages, in a certain way, to adjust equity and incentives according to the state of the resource fund. This helps us to explain and understand why, both in Koster et al.[8] and in our results, even after introducing real behavioral heterogeneity, the interpolated mechanism appears again as the most balanced option.

6.2 The role of behavioral composition in the performance of mechanisms

Although the partial confirmation of the main finding of Koster et al.[8] is highly relevant for our study, the most interesting result of our work appears when analyzing the second research question: does the performance of the mechanisms depend on the behavioral composition of the groups? Our results clearly show that it does, which leads us to strongly accept our second hypothesis (H2), since it is confirmed that group composition clearly influences the results, and it does so especially when groups combine very different behavioral profiles.

What our results show, presented in the previous section, is not only that some groups perform better than others, but that the effect of the mechanism depends on the type of players that make up the group. Therefore, we could conclude that it is not enough to ask which mechanism is better, we must also ask for which type of group each mechanism works better.

Regarding homogeneous groups, the scope for action of the mechanism is much smaller. Our results reveal that when all players are highly cooperative, outcomes are high under any redistribution rule, whereas when all are poorly cooperative, outcomes are low regardless of the mechanism. Therefore, in these cases we can affirm that institutional design hardly alters the final outcome obtained.

However, we see that this situation changes considerably when we move to analyze mixed groups and, above all, asymmetric groups, since in these cases the mechanism does begin to make clear differences in the results. This finding fits well with what the reviewed literature on social dilemmas suggests, because when very cooperative individuals coexist with others who barely contribute within the same group, an incentive problem arises, since cooperators sustain the resource but at the same time are exposed to others benefiting. In this type of situation, the problem is no longer only how much is distributed, but how these incentives are managed within the group. That is, the mechanism shifts from being a simple allocation rule to having a more active role, aligning (or not) individual incentives with the

collective interest. For this reason, in more heterogeneous groups, small differences in the design of the mechanism can translate into much larger changes in final outcomes.

For all these reasons, the main conclusion we draw from our results is that mechanisms really make a difference when groups are internally diverse in their behavior. That is, the greater the difference between individuals, the more important the design of the rules becomes. While Koster et al.[8] focus their study on analyzing the design of the mechanism itself, we contribute to their findings by providing empirical evidence on the importance of group composition. In our case, we see that the performance of a mechanism is not fixed, but rather something that also depends largely on the nature of the group of individuals in which each mechanism is tested.

Furthermore, our results lead us to question an idea that sometimes appears implicitly in the literature on this type of social dilemma, which is that heterogeneity is simply an obstacle to cooperation. In our case, heterogeneity does not always produce worse results, but rather makes more visible the importance of the type of mechanism applied. That is, what our results show is that, when there are clear differences between behavioral profiles, institutional design becomes much more decisive.

This finding suggests that it is not enough to distinguish between homogeneous and heterogeneous groups in general terms, but that the specific type of heterogeneity present in the group also matters. The clearest evidence, as we have seen, appears in asymmetric compositions, which are precisely those in which we observe the greatest differences between mechanisms. For example, in the composition "3 High + 1 C0", total surplus increases from 387.2 under Equal to 1429.7 under Proportional and 1689.4 under Interpolating. A similar pattern also appears in other unequal compositions, such as "2 C0 + 2 High", where the differences between mechanisms once again widen notably.

This effect is also reflected particularly clearly in the behavior of the proportional mechanism. In asymmetric groups, where part of the individuals concentrates a large share of contributions, this mechanism not only rewards that behavior, but also cumulatively amplifies the differences between players. As a result, although it maintains high levels of efficiency, it does so at the cost of a much more pronounced increase in inequality compared to the other mechanisms. This pattern does not appear with the same intensity in the interpolated mechanism, which manages to sustain similar levels of surplus without generating such a marked divergence between individuals.

These patterns indicate that, when the group combines highly cooperative profiles with less cooperative ones, the way resources are distributed becomes especially important. In homogeneous groups, inequalities between players barely appear, since all behave similarly and, consequently, obtain similar outcomes. However, in heterogeneous and asymmetric groups, the relationship between efficiency and inequality becomes much more complex, and it is precisely there where we are truly able to observe and test whether a mechanism can provide high performance without triggering inequality.

7 Conclusions

This paper examines the extent to which the results of Koster et al. (2025) [8] hold when introducing behavioral heterogeneity based on real data, as well as to examine the role of group composition in the performance of allocation mechanisms. Based on the results obtained, we have partially confirmed the results of the reviewed paper and, at the same time, provided empirical evidence that such performance is not uniform, but rather depends on group composition.

In this final section, after discussing the main findings of the study, we will explore their implications and point out the main limitations of our analysis, as well as possible lines of research that remain open.

First, the results of our study have highly relevant implications, since they show that it is not enough to analyze mechanisms by themselves, but that it is necessary to take into account how individuals behave and how they are combined within the group. If the performance of a mechanism also depends on group composition, then evaluating allocation rules under assumptions of

group homogeneity may lead to simplified and erroneous conclusions.

Therefore, these findings imply that the capacity of the mechanisms themselves is sometimes not sufficient on its own to generate significant changes and that their functioning largely depends on the nature of the group to which they are applied, since it is not the same to manage a group where everyone behaves similarly as one with very different behavioral profiles. Therefore, it can be concluded that these mechanisms seem to work better when they adapt both to the state of the resource and to the behavioral composition of the group.

However, we also point out some limitations of our study. First, although we introduce behavioral heterogeneity based on real data, we still operate in a simulated environment. This means that the results should not be interpreted as a direct test of how real human groups would respond in direct interaction with these mechanisms, but rather as a more realistic approximation than assuming homogeneous agents.

Second, our measure of sustainability is more limited than in Koster et al., [8] since in our simulations all games reach the maximum number of rounds. For this reason, the comparison between mechanisms is carried out in terms of surplus and inequality, and not in terms of system survival or player persistence.

From this point, we consider that an interesting future line of research could be to study the functioning of mechanisms that explicitly consider the behavioral composition of the group. That is, in our case, the mechanisms remain fixed regardless of the type of group, but our results suggest that their performance varies systematically with composition. This leads us to propose the possibility of continuing to explore whether it is possible to design mechanisms that adjust not only to the state of the resource, but also to the type of individuals that make up the group.

8 References

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